

Multi-institutional Evaluation of Medialization Laryngoplasty in the Elderly

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Abstract

Objective. To evaluate voice outcomes of medialization laryngoplasty in the elderly population (65 years and older) and to identify swallow outcomes, complication rates, and predictors of voice outcomes.

Study Design. Case series with chart review.

Setting. Two tertiary academic medical centers.

Subjects and Methods. We retrospectively reviewed charts of 136 patients age 65 years and older undergoing medialization laryngoplasty between January 2008 and May 2016 at 2 tertiary academic institutions. Primary outcome was assessed using Voice Handicap Index 10 (VHI-10) score and Grade, Roughness, Breathiness, Asthenia, and Strain (GRBAS) score. Secondary outcomes were assessed using the Eating Assessment Tool 10 (EAT-10) when dysphagia was present, stroboscopic analysis of glottic closure, and complication rates. A logistic regression analysis assessed predictors of voice improvement after medialization laryngoplasty.

Results. Total GRBAS and VHI-10 scores showed a significant improvement postoperatively ($P < .05$). A $\geq 20\%$ improvement was seen in 81.6% of patients, and a $\geq 50\%$ improvement was seen in 53.7%. No patient had major complications. Minor complications occurred in 5.9% of patients. Multivariable logistic regression identified preoperative injection augmentation as an independent predictor of less improvement in VHI-10 score ($P = .015$). Voice therapy prior to medialization did not affect voice outcomes ($P = .640$).

Conclusion. Patient- and provider-perceived voice quality are significantly improved after medialization laryngoplasty in the elderly, and the procedure is associated with a low complication rate even in an elderly cohort. Improvement in patient-perceived voice outcomes after medialization laryngoplasty was diminished in patients with preoperative injection augmentation.

Keywords

medialization laryngoplasty, medialization thyroplasty, injection augmentation, vocal fold paralysis, glottic insufficiency, vocal fold atrophy, elderly patients

The proportion of individuals above the age of 65 has increased significantly due to an increase in life expectancy. Data suggest that 20% to 58% of patients older than 65 years have voice complaints, and up to 85% of patients in this population have evidence of glottic insufficiency, irrespective of voice symptoms.^{1–3} Dysphonia in this population is often related to changes in the larynx or to an increased incidence of pathology. The subsequent glottic insufficiency leads to “leaking” of air during phonation, a weakened cough, difficulty swallowing thin liquids, and, in severe cases, an increased risk of aspiration. Glottic insufficiency can be divided into paralytic and nonparalytic causes.

In the case of vocal fold paresis or paralysis, recovery depends on etiology and extent of nerve damage. While awaiting recovery, which may take up to a year, temporary injection augmentation is often performed to improve glottic closure. If recovery does not occur, permanent interventions include medialization laryngoplasty, arytenoid adduction, and laryngeal reinnervation. Medialization laryngoplasty is particularly commonly performed and has advantages of low cost, reversibility, avoidance of general anesthesia, and ability to adjust the implant intraoperatively.^{4,5} Noted complications include edema, extrusion, airway obstruction, and wound complications.^{6–9} Revision rates due to complications or persistent glottic insufficiency vary from 6% to 8%.^{6,9}

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Outcomes of medialization laryngoplasty have been studied extensively in the general adult population.^{4,10} While medialization laryngoplasty has been studied in the elderly population, published work dedicated to outcomes in the elderly population specifically is lacking.¹¹ Elderly patients often question medialization due to their perception that elective surgery should not be pursued at advanced ages, and there is a dearth of literature regarding safety and effectiveness in this group.

The primary objective of this study is to evaluate voice outcomes after medialization laryngoplasty in the elderly population as measured by the patient-perceived Voice Handicap Index 10 (VHI-10) and the physician-perceived Grade, Roughness, Breathiness, Asthenia, and Strain (GRBAS) score. The secondary objectives are to assess swallowing outcomes as measured by the Eating Assessment Tool 10 (EAT-10), assess complication rates, and identify predictors of voice outcome. The clinical goal is to help inform preoperative discussions with elderly patients.

Methods

Study Population

After approval by The Ohio State University Wexner Medical Center Institutional Review Board and The University of Texas Health Science Center at San Antonio Institutional Review Board, a retrospective chart review was conducted on patients older than 65 years who underwent medialization laryngoplasty between January 2008 and May 2016 at 2 tertiary institutions. Patients had at least 1 month of follow-up and were excluded if they were <65 years, had no follow-up, or had a history of unrelated laryngeal surgery.

Patients were confirmed to have glottic insufficiency using clinically assessed videolaryngostroboscopy and were offered medialization laryngoplasty for indications of vocal fold atrophy or paresis/paralysis. A flexible scope was introduced transnasally to view the larynx. Glottic closure ratings were subjectively assessed by experienced surgeons. A medialization laryngoplasty was performed under monitored anesthesia care in the operating room.¹² The implants used were either Gore-Tex (Gore Medical, Flagstaff, Arizona) or silastic (silicone) implants. Proper placement of the implant was assessed using flexible transnasal laryngoscopy and patient's voice quality during the operation.

Dependent and Independent Variables

Charts were reviewed for patient characteristics, including demographic, preoperative, surgical, and postoperative data. To assess voice outcome, the VHI-10 and GRBAS scores were evaluated. The VHI-10 is a patient-determined assessment of voice and associated quality of life. The GRBAS score is a provider-determined assessment of voice quality. The VHI-10 and GRBAS scores are subjective ratings with high reliability and validity.^{13,14} We assessed dysphagia, when present, using the EAT-10 scale, a validated, patient-determined scale used to assess dysphagia and associated quality of life.¹⁵ All surveys were administered pre- and

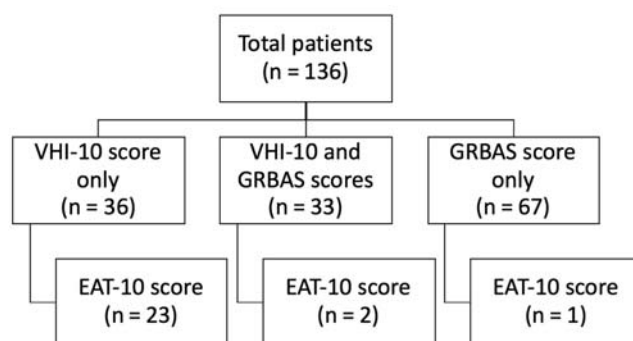


Figure 1. Proportion of patients with voice and swallow outcomes as measured with the Voice Handicap Index 10 (VHI-10); Grade, Roughness, Breathiness, Asthenia, and Strain (GRBAS); and Eating Assessment Tool 10 (EAT-10) scores.

postoperatively. The GRBAS score was assigned by the laryngologist and sometimes also by the speech-language pathologist (SLP) independently.

To assess complication rate, intraoperative and postoperative complications were recorded. Implant extrusion, rapidly expanding hematoma, airway obstruction, and major perioperative cardiopulmonary events were classified as major complications. Minor complications included edema leading to clinical sequelae or requiring treatment, hypertrophic scar formation, wound dehiscence, ecchymosis, small nonobstructive hematoma, and minor injury to surrounding structures.

Because only a portion of the patient cohort had complete data for both VHI-10 and GRBAS score (**Figure 1**), we chose to classify patients in groups of <20%, 20% to 50%, and >50% improvement in voice outcomes. A 50% improvement in voice outcomes was chosen based on our mean percentage improvement in VHI-10 score of 46.6% and mean percentage improvement in GRBAS score of 55.5%, for a mean improvement of 51.9% in both scales. If patients had both preoperative and postoperative VHI-10 and GRBAS scores, >50% improvement in both VHI-10 and GRBAS was necessary to be classified as having a good outcome. Alternatively, if patients only had preoperative and postoperative VHI-10, >50% improvement in VHI-10 was necessary to be classified as having a good outcome. Similarly, if patients only had preoperative and postoperative GRBAS, >50% improvement in GRBAS was necessary to be classified as having a good outcome. To identify predictors of good outcome (>50% improvement in voice, as measured by VHI-10 and GRBAS scores), age, sex, laterality of procedure, perioperative anticoagulants/antiplatelet therapy, implant type (silicone vs Gore-Tex), etiology (paralytic vs nonparalytic), medical comorbidities, preoperative clearance, previous voice therapy, previous injection augmentation, and time between symptom onset and surgery were noted.

Statistical Analysis

Statistical analysis was conducted on SPSS Version 24 (SPSS, Inc, an IBM Company, Chicago, Illinois). Descriptive statistics included means, standard deviations, ranges, 95%

confidence intervals (95% CIs), frequency, and percentages. The preoperative vs postoperative VHI-10, EAT-10, and total GRBAS scores were analyzed using paired *t* tests. The preoperative vs postoperative measure of each component of the GRBAS score was analyzed using a related-samples Wilcoxon signed rank test. A Fisher's exact test was conducted to assess correlation between complication rate and pre- and perioperative antiplatelet/anticoagulant therapy. A χ^2 test was conducted to assess correlation of revision medialization with degree of voice improvement.

A univariable logistic regression was conducted to evaluate these variables as predictors of good outcome. The odds ratio (OR) and 95% CIs of each variable were estimated by the univariable logistic model, and a variable was retained for inclusion based on significance at a .15 α level. A multivariable logistic regression model was applied to evaluate effect on outcome after medialization. All *P* values were 2-sided, and significance was set to .05 α .

Results

Clinical Characteristics

In total, 136 patients aged ≥ 65 years undergoing medialization laryngoplasty between January 2008 and May 2016 met inclusion criteria (**Table 1**). In 86 of 136 (63.2%) cases, patients had glottic insufficiency due to paralytic causes, and in 33 of 136 (24.3%) cases, glottic insufficiency was due to nonparalytic causes. The remaining 17 of 136 (12.5%) cases had a diagnosis of both atrophy and paralysis, of unrelated etiologies, each independently meriting medialization. The breakdown of these etiologies appears in **Table 2**. Of the 86 patients with paralytic glottic insufficiency, 2 of 86 (2.3%) had bilateral medialization; of the 33 patients with nonparalytic glottic insufficiency, 26 of 33 (78%) had bilateral medialization; of the 17 patients with a diagnosis of both atrophy and paralysis, 13 of 17 (76.5%) had bilateral medialization.

In our cohort, 50 of 136 (36.8%) patients were on antiplatelet therapy (**Table 1**); 38 of 50 (76.0%) of these patients were on aspirin only, and 12 of 50 (24.0%) were on aspirin and clopidogrel. In addition, 14 of 136 (10.3%) were on anticoagulants; 10 of 14 (71.4%) of these patients were on coumadin, and the remaining 4 of 14 (28.6%) were on novel anticoagulants (apixaban, rivaroxaban, and dabigatran). Two patients (1.5%) were on coumadin and either clopidogrel or aspirin at the time of surgery. Ultimately, a total of 39 of 136 (28.7%) patients remained on antiplatelet or anticoagulation therapy perioperatively. This was defined as continuing antiplatelet or anticoagulation therapy within 5 days of the operation. In specific, 35 of 136 (25.7%) patients remained on aspirin perioperatively, 3 of 136 (2.21%) of whom also remained on clopidogrel; 3 of 136 (2.21%) patients remained on coumadin perioperatively; and 1 of 136 (0.74%) patients remained on rivaroxaban perioperatively. The remaining patients stopped taking their antiplatelet or anticoagulation therapy at least 5 days preoperatively.

Table 1. Demographics, Comorbidities, Surgical Details, and Etiology of Glottic Insufficiency in the Patient Population (N = 136).^a

Characteristics	Value
Age, mean (95% CI), y	73.2 (72.2-74.2)
Sex	
Male	73 (53.7)
Female	63 (46.3)
Presentation	
Dysphonia	136 (100)
Dysphagia to thin liquids	54 (39.8)
Dyspnea with phonation	30 (22.1)
Comorbidities	
Hypertension	76 (55.9)
History of malignancy	65 (47.8)
CVD	64 (47.1)
Hyperlipidemia	50 (36.8)
Thyroid conditions ^b	35 (25.7)
Diabetes	28 (20.6)
OSA	12 (8.8)
Time between symptom onset and surgery, mean (95% CI), mo	34.4 (24.6-44.1)
Cause	
Paralytic	86 (63.2)
Nonparalytic	33 (24.3)
Combination of both	17 (12.5)
Previous voice therapy	48 (35.3)
Previous injection augmentation	60 (44.1)
Formal preoperative clearance	37 (27.2)
Antithrombotic therapy	
Antiplatelet therapy	50 (36.8)
Anticoagulants	14 (10.3)
Both	2 (1.5)
Perioperative antithrombotic	
Antiplatelet therapy	35 (25.7)
Anticoagulants	4 (2.9)
Both	0 (0.0)
Laterality of procedure	
Left	63 (46.3)
Right	32 (23.5)
Bilateral	41 (30.1)
Implant type	
Gore-Tex	105 (77.2)
Silastic	27 (19.9)
Both	4 (2.9)
Follow-up, mean (95% CI), mo	9.66 (6.96-12.4)

Abbreviations: CI, confidence interval; CVD, cardiovascular disease; OSA, obstructive sleep apnea.

^aValues are presented as number (%) unless otherwise indicated.

^bThyroid conditions include hypo- and hyperthyroidism and exclude thyroid malignancies.

Functional Outcomes

One hundred patients (73.5%) had preoperative and postoperative surgeon-assigned GRBAS scores. Postoperative

Table 2. Etiology of Paralytic and Nonparalytic Glottic Insufficiency (n = 103).^a

Characteristic	No. (%)
Paralytic (n = 103)	
Iatrogenic	52 (50.5)
Idiopathic	23 (22.3)
Malignancy	22 (21.4)
Cerebrovascular accident	4 (3.9)
Trauma	2 (1.9)
Nonparalytic (n = 50)	
Presbylarynx	34 (68.0)
Parkinson's disease	11 (22.0)
Cerebrovascular accident	2 (4.0)
Amyloidosis	2 (4.0)
Amyotrophic lateral sclerosis	1 (2.0)

^aSeventeen patients had a diagnosis of both atrophy and paralysis, of unrelated etiologies, each independently meriting medialization.

total GRBAS score (3.82; 95% CI, 3.33-4.31) was significantly lower than preoperative total GRBAS score (9.02; 95% CI, 8.38-9.66) by 5.20 points (95% CI, 4.63-5.78; $P < .0005$). In addition, there was a significant decrease in value between each specific component of the GRBAS score ($P < .0005$).

In 69 of 136 (50.7%) cases, patients had preoperative and postoperative VHI-10 available for review. Mean postoperative VHI-10 (15.0; 95% CI, 12.0-18.1) was significantly lower than the mean preoperative VHI-10 (29.1; 95% CI, 26.8-31.4) by 13.9 points (95% CI, 10.6-17.2; $P < .0005$).

Of the 54 patients who also complained of dysphagia, 21 (38.9%) patients filled out both preoperative and postoperative EAT-10 scales. There was no significant difference ($P = .063$) in the mean preoperative EAT-10 score (12.3; 95% CI, 7.64-17.1) and the mean postoperative EAT-10 score (7.64; 95% CI, 3.98-11.3).

None of our patients had a major complication. The overall and minor complication rate was 8 of 136 (5.9%) and included 2 of 136 (1.5%) small and nonobstructive hematomas, 1 of 136 (0.7%) persistent vocal fold edemas with clinical sequelae affecting vocal outcome, 1 of 136 (0.7%) wound dehiscences, 1 of 136 (0.7%) hypertrophic scars, 1 of 136 (0.7%) ecchymoses, 1 of 136 (0.7%) injuries to the ventricular mucosa, and 1 of 136 (0.7%) cricothyroid membrane injuries. Minor edema with no clinical significance was present in 23% of patients, was expected, and was not recorded as a complication. One of the patients who developed a small nonobstructive hematoma was on coumadin but had stopped therapy preoperatively. The other patient was on aspirin and clopidogrel and continued both. The patient who developed ecchymosis was on clopidogrel and aspirin; the clopidogrel was discontinued preoperatively, while the aspirin was continued. There was no correlation between overall complication rate and preoperative antiplatelet/anticoagulant therapy ($P = .212$) and perioperative antiplatelet/anticoagulant therapy ($P = .108$).

Patients were divided into groups by degree of improvement of voice scores (<20%, 20%-50%, or $\geq 50\%$ improvement, as detailed above), and **Figure 2** shows the complications that occurred in each group. In 24 of 136 (17.6%) patients, minor vocal fold edema was noted on postoperative stroboscopy but did not require any treatment or have any clinical consequence. Edema was resolved in all these cases within 4 months.

Incomplete glottic closure was noted in 26 of 136 (19.1%) patients postoperatively. Postoperative incomplete glottic closure predicted a <50% improvement in voice scores ($P < .0005$). In 3 of 26 (11.5%) of these cases, patients required revision medialization for poor voice quality from postoperative incomplete glottic closure, for an overall rate of revision due to persistent glottic insufficiency of 2%. An additional 3 patients with complete closure postoperatively had revision medialization for poor voice quality, yielding an overall revision rate of 4%. Of these revision medialization patients, 2 of 6 (33.3%) had $\geq 50\%$ improvement in voice scores, 1 of 6 (16.7%) had 20% to 50% improvement in voice scores, and 3 of 6 (50%) had <20% improvement in voice scores.

Univariable Analysis

A univariable logistic regression model identified the variables of thyroid conditions (OR, 1.955; 95% CI, 0.879-4.349; $P = .100$), cardiovascular disease (OR, 1.744; 95% CI, 0.881-3.451; $P = .111$), previous injection augmentation (OR, 0.367; 95% CI, 0.183-0.738; $P = .005$), and previous voice therapy (OR, 0.538; 95% CI, 0.264-1.097; $P = .088$) as variables to include in a multivariable regression model to predict $\geq 50\%$ improvement in voice scores based on a P value of $< .15 \alpha$ (**Table 3**).

Multivariable Analysis

Although thyroid conditions as a comorbidity produced a P value $< .15 \alpha$ in the univariable analysis, this was removed in the multivariable logistic regression due to hypothyroidism being a known confounder to thyroidectomy. After adjusting for other factors, only previous injection augmentation remained statistically significant (OR, 0.388; 95% CI, 0.180-0.834; $P = .015$) (**Table 4**), showing a negative correlation with achieving $> 50\%$ improvement in VHI-10 and GRBAS scores.

Patients Stratified by Previous Injection Augmentation

We found similar VHI scores before medialization laryngoplasty (and after injection, where applicable) in patients with and without previous injection augmentation (31.1 ± 9.3 vs 27.2 ± 10.7 , $P = .103$), but we found significantly better GRBAS scores before medialization laryngoplasty in the previously injected group (7.47 ± 3.10 vs 9.89 ± 2.96 in the uninjected group, $P < .0005$) (**Table 5**). After medialization laryngoplasty, we found similar GRBAS scores between injected and noninjected groups (3.61 ± 2.28 vs 3.94 ± 2.55 , $P = .525$) but significantly worse VHI scores in the injected group (18.6 ± 13.5 vs 11.5 ± 13.4 , $P =$

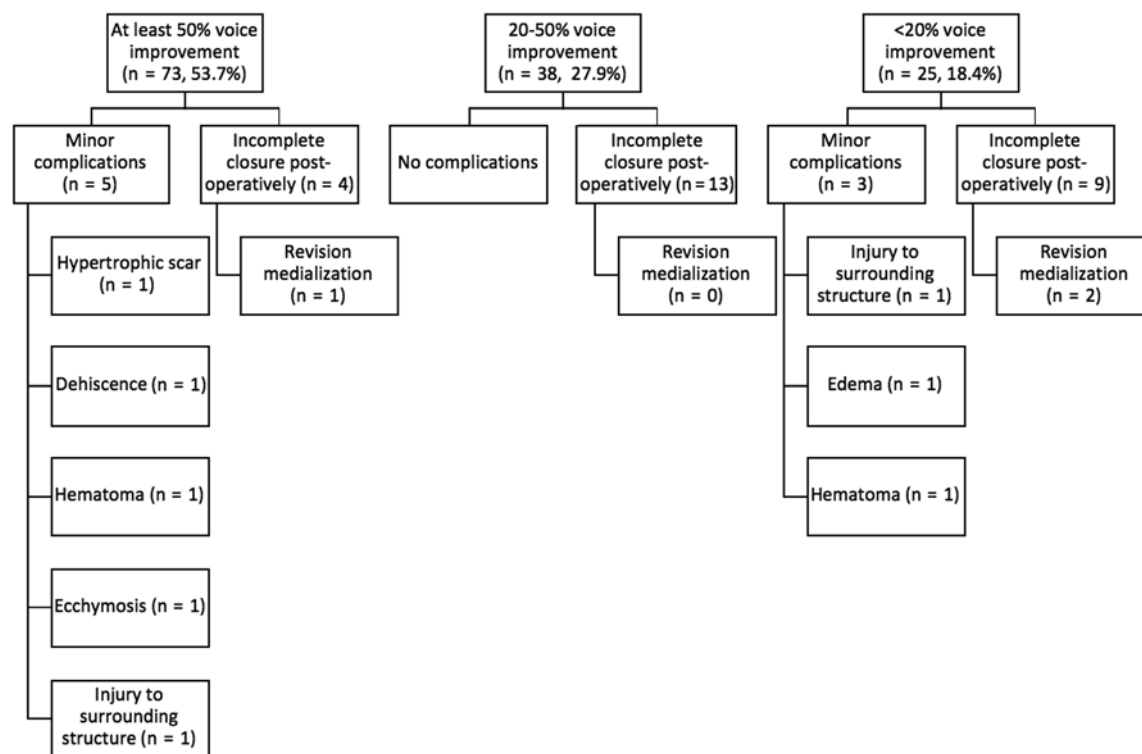


Figure 2. Outcomes using voice improvement as parameter. Voice improvement was assessed by Voice Handicap Index 10 and total Grade, Roughness, Breathiness, Asthenia, and Strain scores when present. Further subdivided by complications, persistent incomplete closure, and need for revision medialization.

.034). **Tables 5** and **6** show further details on these patients.

There was no statistically significant difference in any clinical factors between the group of patients who underwent prior injection augmentation ($n = 60$) who achieved $>50\%$ improvement in voice scores ($n = 24$) vs those who achieved $<50\%$ improvement ($n = 36$).

Discussion

Our study shows significant improvement in voice outcomes as measured by VHI-10 and GRBAS scores. The complication rate of 5.9% included only minor complications. Injection augmentation prior to medialization laryngoplasty was the sole predictor of diminished degree of voice improvement ($P = .015$). The postoperative surgeon-assigned GRBAS scores, however, were similar in patients who had undergone injection as compared to patients who had not.

Many studies have noted the success of medialization laryngoplasty in improving voice in the adult population, with reported ranges of 92% to 100% of patients achieving improvement.^{4-6,9,16} Our elderly population was comparable, with 117 of 136 (86%) improving their postoperative GRBAS and VHI-10 scores. Revision medialization was needed in 6 of 136 (4.4%) of our study sample, all due to poor voice quality, compared to 6% to 8% in the literature on the general adult population.^{6,9}

Dysphagia was a preoperative complaint in 39.8% of our sample, 38.9% of whom completed the EAT-10 scale. We

found no significant improvement in EAT-10 scores in this study. Previous studies have found significant improvement in dysphagia after medialization laryngoplasty as assessed by EAT-10 scores ($P < .01$ in a study of 44 adult patients by Belafsky et al¹⁷). Only 21 patients in our study completed the EAT-10 survey.

Known complications in medialization laryngoplasty include vocal fold edema, implant extrusion, airway obstruction, and wound or bleeding complications; these occur in 6.5% to 32.7% of the adult population.⁶⁻⁹ Our overall complication rate (5.9%) was similar to that of the adult population. Our rate of major complications (0%) was favorable compared to studies on the general adult population (2.6%-8.6%), as was our minor complication rate (5.9%), compared to a range of 3.9% to 24%.⁶⁻⁹

The association between anticoagulants/antiplatelet therapy and postoperative complications is particularly relevant to elderly patients, who are more likely to be on these medications. One series of adult medialization laryngoplasty patients reported that all their complications occurred in patients on antiplatelet or anticoagulation therapy.¹⁸ However, a cohort of elderly patients studied by Bowen et al¹¹ revealed no relationship between intraoperative aspirin use and the incidence of complications ($P > .05$). In our cohort, no patient developed hematoma obstructing the airway. This is notable, given that 48.5% of patients were on antiplatelets/anticoagulants, and only 40.9% stopped them before surgery. There were 2 small hematomas in our

Table 3. Univariable Logistic Regression to Assess Predictors of $\geq 50\%$ Improvement in Voice Outcome as Assessed by GRBAS and VHI-10 Scores.^a

Variable	OR (95% CI)	P Value
Age	0.997 (0.942-1.057)	.931
Sex		
Female	Reference	
Male	1.399 (0.710-2.754)	.332
Time from symptom onset until surgery	0.995 (0.988-1.002)	.168
Causes		
Paralytic	Reference	
Nonparalytic	0.556 (0.247-1.253)	.157
Preoperative clearance	1.778 (0.775-4.128)	.173
Perioperative thrombotic therapy	1.348 (0.632-2.872)	.440
Laterality of procedure		
Left/right	Reference	
Bilateral	0.755 (0.362-1.573)	.452
Implant type		
Silastic	Reference	
Gore-Tex	0.647 (0.271-1.544)	.327
Comorbidities		
Thyroid conditions ^b	1.955 (0.879-4.349)	.100
OSA	1.815 (0.520-6.342)	.350
Hypertension	1.129 (0.567-2.248)	.729
Hyperlipidemia	1.160 (0.575-2.337)	.679
Diabetes	1.435 (0.615-3.348)	.403
Cardiovascular disease	1.744 (0.881-3.451)	.111
Malignancy	1.632 (0.827-3.224)	.158
Previous voice therapy	0.538 (0.264-1.097)	.088
Previous injection augmentation	0.367 (0.183-0.738)	.005

Abbreviations: CI, confidence interval; GRBAS, Grade, Roughness, Breathiness, Asthenia, and Strain; OR, odds ratio; OSA, obstructive sleep apnea; VHI-10, Voice Handicap Index 10.

^an = 136 unless otherwise noted.

^bThyroid conditions include hypo- and hyperthyroidism and exclude thyroid malignancies.

Table 4. Multivariable Logistic Regression Predicting Likelihood of at Least 50% Improvement in Voice Based on Variables from Univariable Analysis with a P Value $< .15 \alpha$.

Variable	OR (95% CI)	P Value
Comorbidity: CVD	1.760 (0.865-3.581)	.119
Previous VT	0.827 (0.373-1.834)	.640
Previous IA	0.388 (0.180-0.834)	.015

Abbreviations: CI, confidence interval; CVD, cardiovascular disease; OR, odds ratio; IA, injection augmentation; VT, voice therapy.

Table 5. VHI-10 and GRBAS Scores Stratified by Patients with and without Previous Injection Augmentation.

Characteristic	Premedialization Scores, Mean (95% CI)	P Value	Postmedialization Scores, Mean (95% CI)	P Value
VHI-10				
No IA (n = 31)	27.2 (23.2-31.1)	.103	11.5 (6.6-16.5)	.034
Previous IA (n = 38)	31.1 (28.1-34.2)		18.6 (14.2-23.0)	
GRBAS				
No IA (n = 64)	9.89 (9.15-10.6)	$< .0005$	3.94 (3.30-4.57)	.525
Previous IA (n = 36)	7.47 (6.42-8.52)		3.61 (2.84-4.38)	

Abbreviations: CI, confidence interval; GRBAS, Grade, Roughness, Breathiness, Asthenia, and Strain; IA, injection augmentation; VHI-10, Voice Handicap Index 10.

Table 6. Clinical Characteristics of Patients Who Underwent Injection Augmentation and Achieved >50% Voice Improvement Postmedialization vs Those Who Achieved <50% Voice Improvement Postmedialization.^a

Characteristic	Previous Injection Augmentation (n = 60)		P Value
	>50% Improvement in Voice Outcomes (n = 24)	<50% Improvement in Voice Outcomes (n = 36)	
Age, mean (SD), y	73.4 (6.47)	72.8 (5.20)	.701
Sex (male)	13 (54.2)	20 (55.6)	1.000
Time from symptom onset until surgery, mean (SD) months	29.4 (39.4)	43.7 (46.1)	.220
Etiology			
Paralytic	16 (66.7)	18 (50.0)	.331
Nonparalytic	5 (20.8)	14 (38.9)	
Both	3 (12.5)	4 (11.1)	
Preoperative clearance	8 (33.3)	9 (25.0)	.564
Laterality of procedure			
Left/right	18 (75.0)	22 (61.1)	.402
Bilateral	6 (25.0)	14 (38.9)	
Implant type			
Silastic	8 (33.3)	7 (20.6)	.364
Gore-Tex	16 (66.7)	27 (79.4)	
Comorbidities			
Thyroid conditions ^b	9 (37.5)	6 (16.7)	.126
Cardiovascular disease	14 (58.3)	14 (38.9)	.189
Malignancy	9 (37.5)	11 (30.6)	.590
Diabetes	4 (16.7)	6 (16.7)	1.000
Hyperlipidemia	11 (45.8)	13 (36.1)	.592
Hypertension	11 (45.8)	19 (52.8)	.792
Obstructive sleep apnea	3 (12.5)	3 (8.3)	.675
Previous VT	12 (50.0)	22 (61.1)	.435
Complication rate	1 (4.2)	1 (2.8)	1.000
Preoperative antithrombotic therapy	8 (33.3)	16 (44.4)	.432
Type of antithrombotic therapy			
Antiplatelet therapy	8 (100)	12 (75.0)	.301
Anticoagulants	0 (0)	3 (18.8)	
Both	0 (0)	1 (6.3)	
Perioperative antithrombotic therapy	5 (20.8)	8 (22.2)	1.000
Length of follow-up, mean (SD) months	12.1 (17.7)	12.2 (15.6)	.995

Abbreviations: SD, standard deviation; VT, voice therapy.

^aValues are presented as number (%) unless otherwise indicated.^bThyroid conditions include hypo- and hyperthyroidism and exclude thyroid malignancies.

series, both in patients on coumadin or clopidogrel. No patient on aspirin alone developed hematoma. There was no statistical correlation between antiplatelet/anticoagulant therapy, stopped preoperatively or not, and either complication rates or voice scores. While caution should be taken in these patients, they can be counseled that antiplatelet/anticoagulant therapy does not clearly predict more complications or worse voice outcomes.

We found that patients without preoperative injection augmentation had a 2.6 odds of achieving $\geq 50\%$ improvement in VHI-10, despite similar postoperative GRBAS scores. The injected patients may be comparing their postoperative result with the often dramatic result they remember from injection augmentation. Because more edema

develops with medialization compared to injection, the voice improvement can be more gradual and less obvious. In turning to the literature, there is no study specifically examining this question. A study by Young et al¹⁹ compared voice outcomes in vocal fold atrophy after injection augmentation vs medialization laryngoplasty. The authors found that 6 of 11 (55%) patients with poor voice outcomes after injection augmentation also did poorly after subsequent medialization laryngoplasty. These data alongside ours highlight that failure to improve significantly after injection augmentation could denote a more difficult patient population. In contrast, a recent study by Merea et al²⁰ found that in 135 adult patients, there was no difference in postoperative VHI-10 score between patients who had injection

augmentation before medialization laryngoplasty (56/135, 41.5%) and patients who had not (79/135, 58.5) ($P = .73$). In our sample, voice outcomes were measured using the GRBAS score in addition to the VHI-10, providing a more comprehensive evaluation of voice outcomes. One additional possibility for the difference in VHI-10 scores postoperatively is that patients may have been medialized when some of the injected filler was still present, leading to underaugmentation and ultimate worsened voice quality. However, the good GRBAS scores at the time of the poor VHI scores point away from this possibility.

Preoperative voice therapy was not associated with improved outcomes in our cohort. Therapy is sometimes used preoperatively to minimize laryngeal hyperfunction and help improve voice outcomes postoperatively, despite a lack of evidence regarding the combined voice therapy and surgical approach to glottic insufficiency.²¹⁻²³ This study provides some evidence that this approach may not be necessary, although it is recognized that a lack of consistency in content, frequency, and type of voice therapy in treatment of glottic insufficiency renders comparison among treatment combinations difficult.

The main limitation in our study is that only a portion of our patient cohort had complete data for both VHI-10 and GRBAS scores, which might have led to sampling bias. We addressed this by classifying patients in groups of <20%, 20% to 50%, and $\geq 50\%$ improvement in outcomes. In addition, preinjection augmentation scores were not available because many patients had injections at outside institutions. Finally, some patients had follow-ups of as little as 1 month, raising the concern that late failure or decrement in benefit over time may not have been captured. This criterion was chosen by the authors because many patients traveled long distances for clinic visits and opted for as-needed follow-up when satisfied with initial outcome; we hoped to avoid excluding patients with positive outcomes who did not require additional visits.

Conclusion

This study demonstrates that medialization laryngoplasty in the elderly leads to improvement in voice symptoms and appears to be free from significant adverse effects. Preoperative injection augmentation was associated with less improvement in patient-perceived voice outcomes after medialization laryngoplasty, even though providers perceived these patients' voices to be similarly improved to noninjected patients. Antiplatelet therapy or anticoagulants did not correlate with increased complication rates or poorer voice or swallow outcomes. These findings will be crucial in guiding the preoperative counseling of the elderly population considering medialization laryngoplasty.

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Author Contributions

Ramez Philips, project design, data acquisition, analysis, and interpretation, manuscript preparation, revision, and approval; **Kevin Chorath**, data acquisition, analysis, and interpretation, manuscript preparation, and approval; **Brad DeSilva**, project design, data acquisition and interpretation, manuscript revision and approval; **L. Arick Forrest**, project design, data acquisition and interpretation, manuscript revision and approval; **C. Blake Simpson**, project design, data acquisition and interpretation, manuscript revision and approval; **Laura Matrka**, project design, data acquisition and interpretation, manuscript revision and approval.

Disclosures

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